

Hasan Malik

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EDUCATION

University of Toronto

B.Sc. in Computer Science; GPA: 3.98/4.00

Toronto, ON

Expected June 2028

EXPERIENCE

Software Engineer (Promoted from Intern)

March 2026 – Present

FlipPilot | Recipient of Best Intern Award

Remote

- Developed industry-specific e-commerce browsing by building a multi-stage **async orchestration pipeline**
- Automated production AI benchmarking across 3 analysis stages via a **parallel 3-LLM pipeline** & dashboard
- Eliminated a 200s pipeline telemetry stall by building a **7-stage** real-time event streaming layer
- Corrected AI valuation anomalies, securing **100% data ingestion fidelity** by adding an enrichment scrape

AI Research Intern

May 2026 – Present

The Matter Lab | Supervised by Alan Aspuru-Guzik

Toronto, ON

- Productionized a **55-package agentic stack** on local & HPC by building a **Docker/Apptainer runtime**
- Accelerated Jetson VLM inference **from 3.7s to 1.2s** by profiling GPU telemetry and tuning power modes

Software Engineer

Dec 2025 – Present

Department of Computer Science, UofT | Supervised by Chandra Gummalur

Toronto, ON

- Built a **client-side rendering engine** enabling viewer-controlled slide navigation independent of the screen-share
- **Synchronized** live presenter annotations across distributed sessions through a **debounced, stateful** event layer
- Shipped a **split-view architecture** rendering annotation and media assets by building parallel graphics pipelines

PROJECTS

TransitSim | *Python, JavaScript, FastAPI, NumPy, React*

- Simulated Toronto transit across **620K** daily trips & **11** climate/GDP metrics via a parameterized 3D digital twin
- Cut simulated CO₂ by **72%** under a 26min commute cap by solving a multi-modal **convex optimization** problem
- Shipped a public API exposing a **6-mode simulation** for developer use on a **containerized** cloud runtime
- Prevented Python/JS engine drift to a 10⁻⁹ relative tolerance by building a cross-language parity test harness

TensorFlow Probability | *Open-Source Contributor | Python*

- Expanded BFGS optimization engine capabilities by engineering **configurable line-search** tuning parameters.
- Validated BFGS parameter forwarding **across 2 optimization trajectories** using Rosenbrock convergence tests.

UC BorrowBox | *Railway, Node.js, Express, PostgreSQL, React*

- Reduced duplicate-purchase waste **across 3 residence halls** via a peer-to-peer exchange with 3 transaction types
- Standardized an 8-view crossplatform UI by codifying Apple's 11-step type scale and color system into tokens

StarTooth | *Overall 3rd, SmileHacks UofT | TypeScript, PostgreSQL, Next.js, React*

- Synthesized interactive 3D dental anatomy from 2D x-ray inputs via a client-side graphics rendering pipeline
- Simulated clinical pathologies and treatments e.g. extractions via real-time 3D mesh modifications
- Facilitated post-appointment consultation via **role-based Q&A** channels and **persistent 3D annotations**

Sparky | *Winner, Spark Case Competition × Hackathon, Western University | SwiftUI, SFSSpeechRecognizer*

- Coordinated medical transit in low-infrastructure regions by engineering a conversational **iOS logistics agent**
- **Eliminated LLM medical liability** by parsing vocal inputs into deterministic synonym-matched intents
- Streamlined patient-to-driver matching by designing a **priority algorithm** for scarce transportation assets

Insight | *MacHacks | SwiftUI, Swift, AVSpeechFoundation*

- Engineered an iOS **conversational agent** for blind users by running **VLM inference** at **1Hz on scene-change**
- Enabled hands-free GPS navigation with **four phases of haptic cues** by polling waypoint distance every **2s**

TECHNICAL SKILLS

Languages: Python, C/C++, Java, Swift, SQL, JavaScript/TypeScript, Assembly (MIPS), Lua, HTML/CSS

Frameworks & Libraries: React, SwiftUI, Node.js, Express, Flask, PydanticAI, Hono, TanStack Query

Databases & ORMs: PostgreSQL, Supabase, Drizzle, Zod

Tools & Infrastructure: GitHub Actions, Docker, Temporal, Railway